THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 10: **How Quantities Combine: Understanding Data** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A group of 20 students has an average of 55 marks and another group of 30 students has an average of 65 marks. The average of all 50 students is: 1

(a) 63 **(b)** 61 **(c)** 62 **(d)** 60

2. A group of 25 students has an average of 90 marks and another group of 15 students has an average of 50 marks. The average of all 40 students is: 1

(a) 74 **(b)** 76 **(c)** 75 **(d)** 77

3. 400 mL of a solution with 5% salt is mixed with 100 mL of a solution with 10% salt. The concentration of salt in the mixture is: 1

(a) 7% **(b)** 6% **(c)** 5% **(d)** 8%

4. 150 mL of a solution with 30% salt is mixed with 350 mL of a solution with 10% salt. The concentration of salt in the mixture is: 1

(a) 18% **(b)** 16% **(c)** 15% **(d)** 17%

5. 6 units of a mixture with 4% metal is mixed with a mixture of 10% metal to get 8% metal. The quantity of the second mixture needed is (same units): 1

(a) 52 **(b)** 12 **(c)** 32 **(d)** 24

6. Marks of 66, 72, 78 are combined with weights 3 : 2 : 5. The weighted average is: 1

(a) 73.2 **(b)** 75.2 **(c)** 74.2 **(d)** 72.2

7. The average of 10 values is 40. When one more value, 62, is added, the new average is: 1

(a) 42 **(b)** 43 **(c)** 44 **(d)** 41

8. How much water must be added to 9 litres of a 20% solution to reduce its concentration to 15%? 1

(a) 5 litres **(b)** 6 litres **(c)** 4 litres **(d)** 3 litres

9. For two groups of equal size, the average of the combined group equals: 1

(a) the simple average of the two group averages
(b) the larger average
(c) the smaller average
(d) their product

10. **Assertion (A):** The average of the averages of two groups of different sizes is not the average of the combined group. 1

Reason (R): The larger group should contribute more to the combined average than the smaller group.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why can the average of the weekly averages be used when every week has the same number of days, but not otherwise? 2

12. A bowl of 3 litres has 6% juice concentrate. How many litres of concentrate does it contain? 2

13. The average of 9 numbers is 40. A tenth number 50 is added. Find the new average. 2

14. State the formula for the combined average when collections of sizes n and m have averages a and b. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Equal quantities of three fertilisers containing $\frac{1}{5}$, $\frac{3}{10}$ and $\frac{1}{2}$ nitrogen are mixed. What fraction of the mixture is nitrogen? 3

16. A goldsmith melts 6 units of 12-varṇa gold with 4 units of 14-varṇa gold. Find the purity of the combined gold (varṇa). 3

17. A bird flew an average of 40 km per day for 20 days and then 61 km on the 21st day. Make a guess, then find the 21-day average. 3

OR

A panipuri vendor has 12 litres of 10% spiced water. How much plain water should be added to bring it to 8%? 3

18. A table shows the number of students and their average heights for 3 classes: (20, 150 cm), (30, 154 cm), (50, 158 cm). Find the combined average height. 3

OR

The average salary of 10 employees is ₹30000. If the manager's salary of ₹52000 is included, find the new average. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In a sports test, strength, speed and skill are combined in the ratio 2 : 3 : 5. Aman scores 80, 70, 60 and Bela scores 60, 70, 80. 5

(i) Without calculating, who is likely to have the higher total and why?

(ii) Find their weighted marks.

(iii) Who wins?

OR

A zoo has 50 deer with average weight 40 kg; males average 46 kg and females 36 kg. 5

(i) Form the equation for the number of males x.

(ii) Find the number of males and females.

(iii) A female deer of 42 kg joins the zoo. Find the new average weight of the females.

20. **Case study:** The table shows weekly spend (₹) of two families. 5

	Housing	Food	Education
Family P	1800	1200	1000
Family Q	900	1500	600

(i) Total spend of each.

(ii) Percentage share of food for each.

(iii) Which chart shows the shares and which the totals?

OR

Case study: A 100% stacked bar shows sales of two bakeries — Bakery A: bread 45%, cakes 35%, snacks 20%; Bakery B: bread 30%, cakes 30%, snacks 40%. 5

(i) Which statements can be inferred:

(a) A sold more bread than cakes;

(b) A sold more bread than B?

(ii) Give two possible sets of totals showing

(b) can be true or false.

END OF QUESTION PAPER — Total: 40 Marks